

Overleaf in 15 Minutes

Writing your lab report in L^AT_EX

COMP 395 – Deep Learning

Fall 2026

Overleaf is a free, browser-based L^AT_EX editor: you type plain text with a few commands on the left, and a typeset PDF appears on the right. Nothing to install. This guide covers only what you need for the short written report: starting a project, headings, images, tables, a little math, and exporting the `.tex` and `.pdf` files you will submit. A complete worked example is in the `sample_report/` folder of your lab repository.

1 Start a project

1. Go to <https://www.overleaf.com> and register (your Oxy email works; the free plan is enough).
2. Click **New Project** → **Blank Project** and give it a name such as `lab2-report`.
3. You now see three panels: the **file list** (left), the **editor** (middle), and the **PDF preview** (right). Your document lives in `main.tex`.
4. Press **Recompile** (or Ctrl/Cmd+Enter) whenever you want to refresh the PDF.

Shortcut: instead of a blank project, choose **New Project** → **Upload Project** and upload a zip of the `sample_report/` folder. Then edit the sample into your own report.

2 The smallest useful document

Delete whatever is in `main.tex`, paste this, and recompile:

```
\documentclass[11pt]{article}

\usepackage[margin=1in]{geometry} % page margins
\usepackage{graphicx}             % images
\usepackage{booktabs}             % nice tables

\title{Lab 2 Report}
\author{Your Name}
\date{Fall 2026}

\begin{document}
\maketitle

\section*{Methods}
I trained a single sigmoid neuron with gradient descent.
```

```

\section*{Results}
My model reached 97\% test accuracy.

\end{document}

```

Things to notice:

- Everything before `\begin{document}` is the *preamble*: settings and packages. Your text goes after it.
- `\section*{...}` makes a heading. Without the `*`, headings are numbered.
- A **blank line** starts a new paragraph. Single line breaks are ignored.
- `%` starts a comment, so a literal percent sign is written `\%`. The same goes for `\_`, `\&`, `\#` and `\$`.
- `\textbf{bold}`, `\emph{italic}`, and `\texttt{code}` style text.

3 Insert an image

1. In the file list, click the **Upload** icon and choose your image. The lab code saves one called `training_results.png`. Avoid spaces in file names.
2. Make sure `\usepackage{graphicx}` is in the preamble.
3. Put this where the figure should go:

```

\begin{figure}[h]
  \centering
  \includegraphics[width=0.8\textwidth]{training_results.png}
  \caption{Training loss (left) and learned weights (right).}
  \label{fig:training}
\end{figure}

```

- `width=0.8\textwidth` means 80% of the text width. Change the number to resize.
- `[h]` asks L^AT_EX to place the figure “here.” It may still move the figure to fit the page, which is normal. That is why you refer to figures by number, not as “the figure below.”
- Refer to it in your text with `Figure~\ref{fig:training}`. L^AT_EX fills in the number. If you see “??”, recompile once more.

4 Insert a table

Columns are separated by `&` and each row ends with `\\`. The `lcc` argument means three columns: left-aligned, centered, centered.

```

\begin{table}[h]
  \centering
  \caption{Accuracy of each model.}
  \label{tab:results}
  \begin{tabular}{lcc}

```

```

\toprule
Model          & Train & Test  \\
\midrule
Sigmoid neuron & 0.987 & 0.974 \\
Random forest  & 1.000 & 0.965 \\
\bottomrule
\end{tabular}
\end{table}

```

which produces:

Table 1: Accuracy of each model.

Model	Train	Test
Sigmoid neuron	0.987	0.974
Random forest	1.000	0.965

- `\toprule`, `\midrule` and `\bottomrule` come from the `booktabs` package. Skip vertical lines; tables read better without them.
- Table captions go *above* the table; figure captions go *below* the figure.
- Prefer clicking? Switch the editor to **Visual Editor** and use **Insert** → **Table**, or build the table at <https://www.tablesgenerator.com> and paste the result.

5 A little math

Inline math goes between dollar signs: `\alpha = 0.01` gives $\alpha = 0.01$. Displayed equations go between `\[` and `\]`:

```

\[
\hat{y} = \sigma(w \cdot x + b) = \frac{1}{1 + e^{-(w \cdot x + b)}}
\]

```

$$\hat{y} = \sigma(w \cdot x + b) = \frac{1}{1 + e^{-(w \cdot x + b)}}$$

Useful pieces: `x^2` (x^2), `w_i` (w_i), `\frac{a}{b}` ($\frac{a}{b}$), `\partial` (∂), `\times` ($\times$), `\hat{y}` ($\hat{y}$).

6 When it will not compile

A red badge next to **Recompile** means errors. Click it to see the log; clicking an error jumps to the line. The usual suspects:

Message	Likely cause
Missing \$ inserted	An unescaped <code>_</code> or <code>^</code> outside math; write <code>_</code>
File 'x.png' not found	Image not uploaded, or the name differs (case matters)
Undefined control sequence	Typo in a command, or a missing <code>\usepackage</code>
Runaway argument	A missing closing brace <code>}</code>
Text vanishes after a %	<code>%</code> started a comment; write <code>\%</code>

Fix the *first* error and recompile; later errors are often side effects of the first.

7 Export and submit

1. Click the **Download PDF** icon above the preview to get the PDF.
2. Open **Menu** (top left) → **Download** → **Source** to get a zip containing your `.tex` file and images.
3. Add the `.tex`, the `.pdf`, and your image files to your lab repository, then commit and push.

Remember the course AI policy: you may use an LLM to help with L^AT_EX boilerplate or debugging, but include the transcript and be ready to explain your report verbally.